

CSCE 632 ACCESSIBLE COMPUTING · MODULE 10 · LECTURE NOTES

Non-Visual Data

Figures, tables, and the hardest requirement in the course.

MODULE OUTCOMES

- Explain why a figure is not an image for this purpose.
- Write a description that conveys what a figure shows.
- Make a data table non-visually navigable.
- Explain sonification and tactile alternatives.
- Make one of your own figures non-visually available.

1 The problem

A FIGURE'S CONTENT IS A RELATIONSHIP, AND ALT TEXT FOR AN IMAGE DOES NOT CONVEY A RELATIONSHIP

'Bar chart of sales by region' says what the image is and nothing whatever about what it shows — which is the characteristic failure of figure alternatives and is extremely common, including in otherwise careful publications.

And the figure's actual point is a comparison, a trend, or an outlier (CSCE 679 Module 01 §2's task framing: the figure exists to support a specific judgement) — so the alternative has to carry the finding, not the picture.

Which means this is a writing problem rather than a markup problem, and is the only requirement in this course that cannot be met mechanically: no tool can generate it, and no tool can check it (Module 09 §3's alt-text example).

So the question to answer is: what would you say if you were describing this figure to somebody over the telephone? — which is the right prompt, and produces usable descriptions almost immediately because it forces you to lead with the point rather than the apparatus.

2 The three layers

- 1 THE SHORT ALTERNATIVE (alt, or a caption)
what kind of figure, what variables, and
THE FINDING in one sentence
"Sales rose in all four regions, with the
north roughly doubling"
- 2 THE LONGER DESCRIPTION (nearby, in real text)
the axes with their units and their ranges
the normalisation (CSCE 679 Module 13)
the shape: trend, groups, outliers
anything a reader would ask about
- 3 THE DATA ITSELF (a real table, or a download)
because the figure is a view of a table
and the table is navigable, searchable, and
exact

AND LAYER 3 IS THE ONE PEOPLE SKIP AND THE ONE THAT
MOST OFTEN SUFFICES -- it costs a markup table, and
it makes every single value available.

AND A REAL DATA TABLE IS NAVIGABLE IN A WAY A FIGURE NEVER IS

A screen reader navigates a properly marked table cell by cell, announcing the relevant row and column headers for each cell — which is a genuinely good interface for exact values, and better than a figure for lookup (CSCE 679 Module 01 §3's division of labour: tables for values, figures for shape).

But it requires the markup: header cells declared as `th`, scope stated, a caption, and no tables used for layout — and a 'table' built from `div` elements is announced as nothing at all (Module 08 §1).

And multi-level or merged headers need explicit association via headers and `id`, which is where table markup gets genuinely difficult and is frequently got wrong — so prefer a flat table, repeated if necessary, over a clever nested one.

Which gives a useful default: publish the figure and the table together — sighted readers use the figure for shape and everybody uses the table for values, which is a curb cut (Module 01 §2) and costs almost nothing since you already have the data.

3 Other channels

- **Sonification maps a data value to pitch**, playing a series as a tune — which **conveys trend and overall shape well and exact values badly** — so it **complements a table rather than replacing one**.
- **And it has exactly the same channel-ranking problem as colour does** — **pitch is a magnitude channel with poor accuracy** — so **CSCE 679 Module 02's argument applies to it directly**, which is a pleasing and slightly deflating result: the auditory channels have a ranking too, and pitch is not near the top of it.
- **Tactile graphics and refreshable braille displays work well for the people who have them and require equipment most readers do not have** — which **limits them to contexts where the equipment is provided**, such as education, and makes them a poor general answer for published material.
- **And structured exploration beats both** — **letting a user query the data ('which is highest', 'what is the value for this category', 'move to the next peak')** is frequently more useful than any rendering of it, and is what the better accessible-chart libraries now do.
- **So the honest ordering is: table, description, structured exploration, sonification**, in decreasing generality. **Structured exploration beats sonification** — a user who can ask 'which is highest' has a **better interface than one listening to a tune**, and it is also easier to build.

4 Practice

EVERY FIGURE GETS A ONE-SENTENCE FINDING AND A TABLE, AND THAT COVERS MOST OF IT

The one-sentence finding is the alternative that matters — and writing it improves the figure's own caption, because it forces you to say what the figure is for (CSCE 679 Module 13 §2's five-clause caption, of which this is the first clause).

And the table makes every value available — which is exact, navigable, searchable, and cheaper than any other option, and is what you already have since the figure was rendered from it.

Which is the curb cut argument again: a figure with a stated finding and a published table is a better figure for everybody — searchable, quotable, re-analysable, and checkable — which is why this requirement is easy to defend.

So this module's access requirement is also CSCE 679's advice — state what the figure shows and publish what it was made from — which is the most satisfying convergence in the semester and the clearest case of the design-for-the-constraint claim being literally true.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
The WAI complex images tutorial (free)	§§1 and 2 — the short-plus-long structure, with worked examples of good and bad descriptions.
The WAI tables tutorial (free)	§2's callout — header association, scope, and the multi-level cases, which are the difficult part.
The Diagram Center image description guidelines (free)	§1 and §2 in depth — by figure type, from people who do this professionally for educational material.
Highcharts and the accessible-charts literature (free docs)	§3's structured exploration — a working implementation of querying a chart rather than rendering it.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Find five published figures** with alt text that describes the image rather than the finding.
- 2 **Describe one figure over the telephone**, and transcribe what you said.
- 3 **Write all three layers** for one of your own figures.
- 4 **Publish the underlying table** for one figure you have made.
- 5 **Navigate a well-marked table with a screen reader**, cell by cell.
- 6 **Build a table from divs** and listen to what is announced.
- 7 **Find a merged-header table** and check its associations.
- 8 **Sonify one series** and try to read an exact value from it.
- 9 **Build a query interface** for one dataset: highest, lowest, by category.
- 10 **Compare all four alternatives** on the same data, and rank them for one task.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why is alt text insufficient for a figure?
- 2 What must the alternative carry?
- 3 Why is this a writing problem, and what prompt helps?
- 4 Give the three layers and what each contains.
- 5 Which layer is skipped, and why does it often suffice?
- 6 What does a table require in markup, and what breaks it?

- 7 What does sonification convey well and badly?
- 8 Why does the channel-ranking argument apply to pitch?
- 9 State the honest ordering of the alternatives.
- 10 Why is this requirement also good visualisation advice?